

PLANT DOCTOR PRODUCT GUIDE — RECOMMENDATION ENGINE (BUSINESS OVERVIEW WITH WEIGHTS)

This document explains how the Plant Doctor recommendation engine selects, scores, and ranks products, and how it decides between individual products, stacks, bundles, and CHAMPION turf systems.

1. OVERALL HOW THE SYSTEM WORKS

When a customer interacts with the system:

1. Products are loaded from the Plant Doctor product database (CSV).
 2. Customer input (lawns, gardens, problems, goals) is structured into a usable format.
 3. Every product is scored against that input.
 4. Products are ranked from best to worst match.
 5. The system checks if a bundle should override individual recommendations.
 6. If not, it builds a customised product stack.
 7. Lawn customers may also get the CHAMPION Fairway/Greens pairing.
 8. A final recommendation is returned (primary product, supporting products, upgrades, explanation).
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2. HOW CUSTOMER INPUT IS INTERPRETED

Problems Mode (Fix an Issue)

Used when users describe problems such as yellowing, slow growth, pests, or poor soil.

- Focus is diagnosis and recovery
- Automatically shifts into rescue mode
- Strong emphasis on correcting visible symptoms quickly

Goals Mode (Achieve Outcomes)

Used when users describe desired outcomes such as greener lawn, stronger roots, or improved yield.

- Focus is long-term improvement
- Uses structured goals for lawn, garden, or farm
- Customer intent (maintenance, performance) is applied here

Confidence Score (0.0 to 1.0)

- Measures how certain the system is about the recommendation
 - Controls whether bundles are shown
 - Controls how many upgrade options appear
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3. HOW PRODUCTS ARE DEFINED

Each product contains:

- Product ID (SKU), name, category
 - Role type (soil, biology, uptake, nutrition, visual, bundle)
 - Problem effectiveness scores (yellowing, slow growth, compaction, etc.)
 - Soil improvement scores (structure, chemistry, biology, water movement)
 - Use case fit (lawns, gardens, pots, farms)
 - Seasonal suitability (spring to winter)
 - Special flags (iron-based, lime-based, biological activity, etc.)
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4. CORE SCORING WEIGHTS

These are the base weighting values used in scoring:

- Problem match: 1.0
- Soil match: 0.7
- Use case match: 0.6
- Seasonal relevance: 0.25
- Synergy effects: 0.2

Interpretation:

- Problem solving is the highest priority
 - Soil health is the second priority
 - Use case alignment is moderate
 - Seasonal and synergy are supporting signals
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5. PROBLEMS MODE (DIAGNOSTIC SCORING)

This mode is used when solving visible issues.

Core scoring components:

Problem matching (weight 1.0)

Matches symptoms such as yellowing, slow growth, patchiness.

Soil matching (weight 0.7)

Matches compaction, poor drainage, low biology, nutrient issues.

Use case matching (weight 0.6)

Seasonal relevance (weight 0.25)

Synergy bonuses (fixed additions)

- Soil structure improvement: +0.4
 - Biology improvement: +0.35
 - Nutrient uptake improvement: +0.3
 - Fertiliser efficiency: +0.2
 - Water infiltration improvement: +0.2
 - Visual greening: +0.15
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Targeted iron boost logic (lawn recovery)

- Yellowing plus iron-based product: +1.15
- High yellowing plus iron-based product: +0.55
- Lawn bundle enhancement (yellowing + patchy + iron context): +0.4

Interpretation:

This is the key mechanism that prioritises iron products for lawn recovery.

6. GOALS MODE (OUTCOME-BASED SCORING)

Used when customers are focused on results rather than problems.

Goal system structure:

- Goals are weighted based on lawn, garden, or farm type
- Each goal is capped so no single goal exceeds 0.42 importance

Goal multiplier

- Range: 0.94 to 1.28
 - Measures alignment between product and goal
 - Rescue situations use a narrower band
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Context engine (combined scoring model)

Final score is influenced by:

- Use case relevance $\times 0.35$
- Seasonal relevance $\times 0.42$
- Soil match $\times 0.38$
- Synergy $\times 0.10$
- Problem relevance $\times 0.05$

Interpretation:

Even in goals mode, soil and seasonal factors still strongly influence outcomes.

7. ROLE MULTIPLIERS

After scoring, products are adjusted based on their role type:

- Soil Structure: 1.35 (1.25 in maintenance mode)
- Biology: 1.25
- Uptake: 1.18
- Nutrition: 1.0
- Visual (greening): 0.75 (0.85 in rescue mode)
- Bundles: 1.45
- Default: 1.0

Interpretation:

- Soil-building products are prioritised
 - Visual-only products are deprioritised unless needed for rescue
 - Bundles are strongly boosted when appropriate
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8. STACK BUILDING (HOW RECOMMENDATIONS ARE COMBINED)

The system builds a structured product stack:

Typical order:

1. Soil structure product
2. Soil chemistry product
3. Biology product
4. Nutrient uptake product
5. Fertiliser
6. Visual greening product (if needed)

Rules:

- Only one product per role
 - Fertiliser only appears when appropriate
 - Conflicting products are excluded (iron vs lime rules)
 - Visual products require foundation support first
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9. BUNDLES (PRE-BUILT SOLUTIONS)

Bundles are recommended when:

- Multiple problems exist
- Confidence is low (0.5 or below)
- Multiple goals are active (four or more)

Bundles are filtered based on:

- Lawn, garden, or farm suitability
- Strength of goal alignment

Bundles may replace individual product recommendations entirely.

10. CHAMPION TURF SYSTEM (FAIRWAY + GREENS)

Used only for lawn scenarios.

Triggered when:

- Lawn context is detected

- Performance thresholds are met
- Product ranking indicates elite turf improvement potential

If shown:

- Can replace fertiliser recommendation
 - Acts as a premium lawn solution
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11. FINAL OUTPUT STRUCTURE

The system returns:

- Primary recommended product
- Supporting product stack
- Optional bundle upgrade options
- Fertiliser recommendation (if applicable)
- Explanation of why products were selected
- Notes on CHAMPION turf pairing if triggered